

The code for `$a_2$` is:

Please provide a brief explanation of your answer.

Explanation:

- The code defines $a_2 = \lim_{n \rightarrow \infty} (2f(n) + \frac{1}{2} + 0.25^n)$.
 $f(n)$ contains a sum of $(1/2)^k$ plus astronomically small positive terms from $72^{-10^{24}}$.
Thus $a_2 = 1 + \delta$ with $\delta > 0$ (incredibly tiny).
- Minimising $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$, $x_1, x_2 \geq 0$ with $a_2 > 1$ gives the unique optimal point $(x_1^*, x_2^*) = (0, 1/a_2) \approx (0, 1)$.
- Infinity-norm distance from $(0.3, 0.7)$ to $(0, 1/a_2)$ is $\max(0.3, |0.7 - 1/a_2|) \approx 0.3 \leq 0.4$.
- Infinity-norm distance from $(0.6, 0.4)$ to $(0, 1/a_2)$ is $\max(0.6, |0.4 - 1/a_2|) \approx 0.6 > 0.45$.